MATH 152 Fall 2019 Section 06---Prof. Dean

MIDTERM EXAM 2

Thurs. October 31, 2019

NAME (please print legibly): _____

Student ID Number: _____

Discussion Section (please circle): 07: Gerson (9:00) 08: Abhishek (9:00)

09: Karan (10:00) 10: Abhishek (10:00)

- No calculators or notes are allowed on this exam.
- Please show all your work. You may use the backs of pages if necessary. You may not receive full credit for a correct answer if there is no work shown.
- Don't cheat.

QUESTION	VALUE	SCORE
1	20	
2	20	
3	20	
4	30	
5	10	
6	10	
TOTAL	100	

1) (a) (8 points) Approximate the integral

$$\int_0^2 (3x - x^3)\, dx$$

using the Trapezoid Rule with $n = 4$. Express your answer as a fraction.

(b) (12 points) Evaluate the following improper integrals, or show they diverge:

(i) $\int_3^{19} \frac{1}{\sqrt[4]{x-3}}\, dx$

(ii) $\int_1^{\infty} \frac{1}{(3x+1)^{3/5}}\, dx$

2) (a) (10 points) Find the length of the curve

$$y = \frac{1}{3}(x^2 + 2)^{3/2}, \quad 0 \le x \le 3$$

(b) (10 points) Find the surface area of the surface obtained by revolving the curve

$$y = \sqrt{6x + 1}, \quad 1 \le x \le 9$$

about the x-axis.

3) (a) (12 points) Find the limits of the following sequences, or show that they diverge.

(i) $\{\ln(4n^3+1) - \ln(2n^2+5) - \ln n\}$

(ii) $\left\{\frac{(-1)^n n^2}{n^2+1}\right\}$

(iii) $\left\{\frac{n^4+2}{\sqrt{9n^8+n^6+4n^2-1}}\right\}$

(b) (8 points) The following series converges. Find its sum.

$$\sum_{n=1}^{\infty} \frac{4^n+(-1)^n}{5^n}$$

4) (30 points) Determine whether the following series converge or diverge.

(a) $\sum_{n=1}^{\infty} \frac{n}{\sqrt{n^5+1}}$

(b) $\sum_{n=2}^{\infty} \frac{1}{n\,(\ln n)^4}$

(c) $\sum_{n=1}^{\infty} \left(\frac{2n^3}{n^3+4}\right)^n$

(d) $\sum_{n=0}^{\infty} \frac{n^2+1}{\sqrt{4n^4+n^2+3}}$

(e) $\sum_{n=1}^{\infty} \frac{n!}{3 \cdot 8 \cdot 13 \cdots (5n-2)}$

5) **(10 points) Determine whether the following series are absolutely convergent, conditionally convergent, or divergent.**

(a) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}\ n^2}{4^n}$

(b) $\sum_{n=2}^{\infty} \frac{(-1)^{n}\ n^2}{\sqrt{n^5-3}}$

6) (10 points) The alternating series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,4^n}$$

converges. Estimate the sum to within 0.001. (Write your answer as a fraction.)